

EMG Stimulation Artefact Suppression

Benchmark Results Report

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Source: emg_3conditions_20260604_110457.xlsx | Conditions: ['Static', 'Moderate', 'Stress'] | Algos: 9

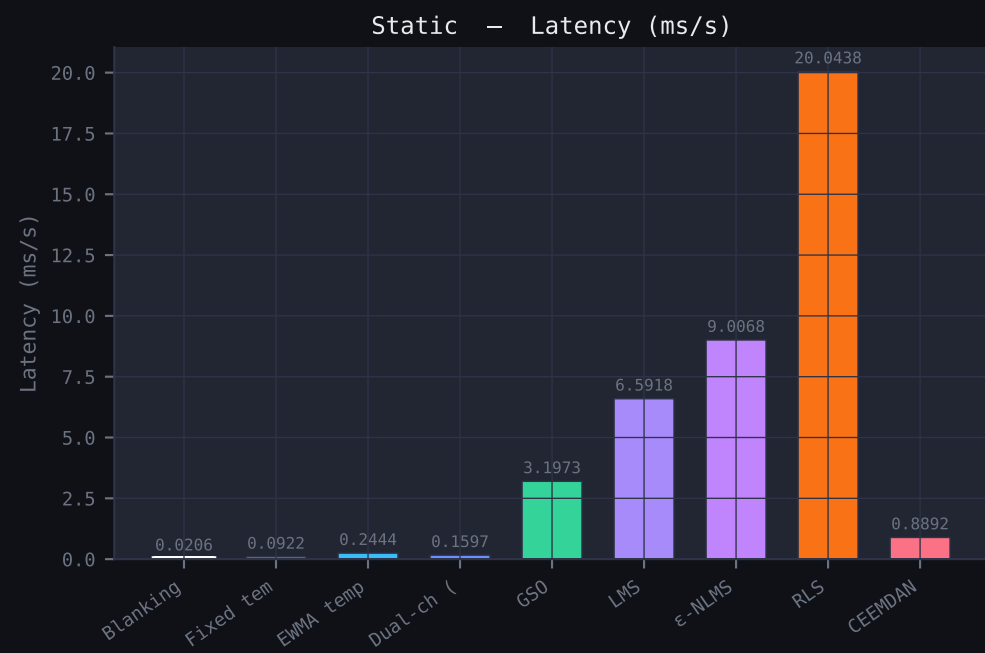
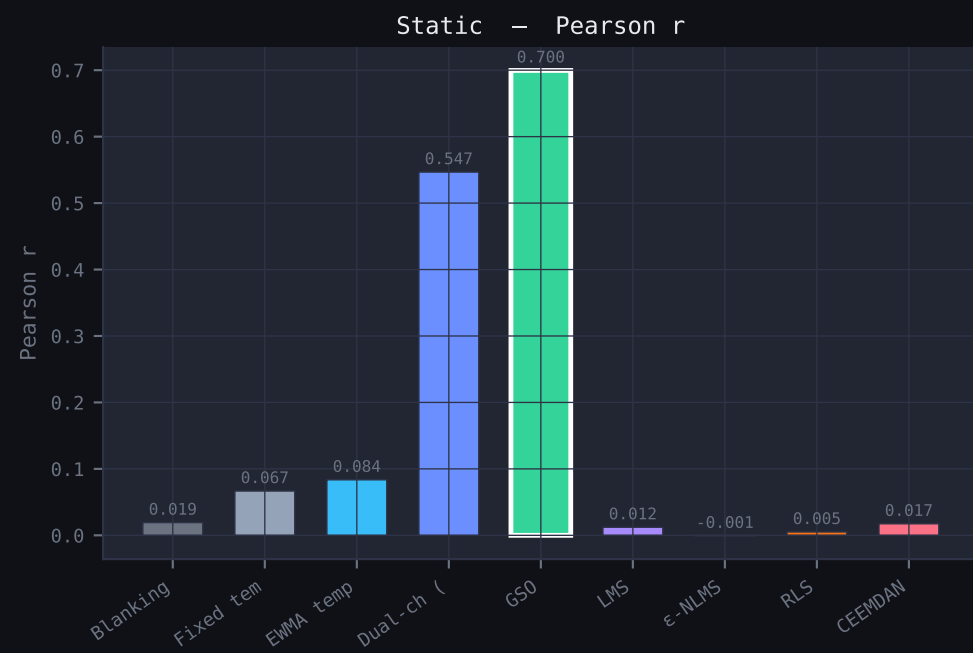
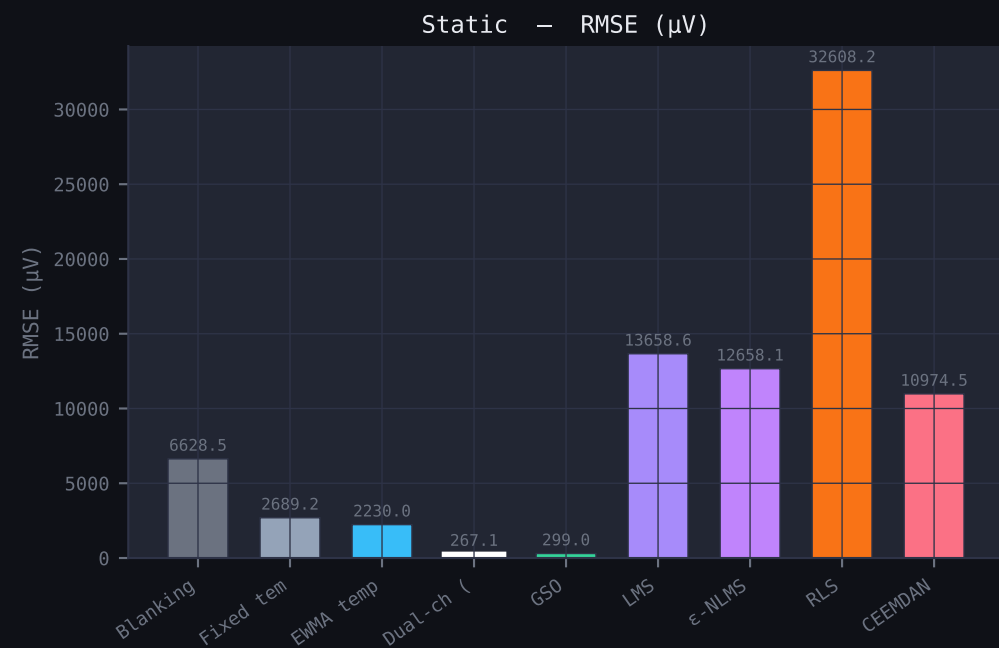
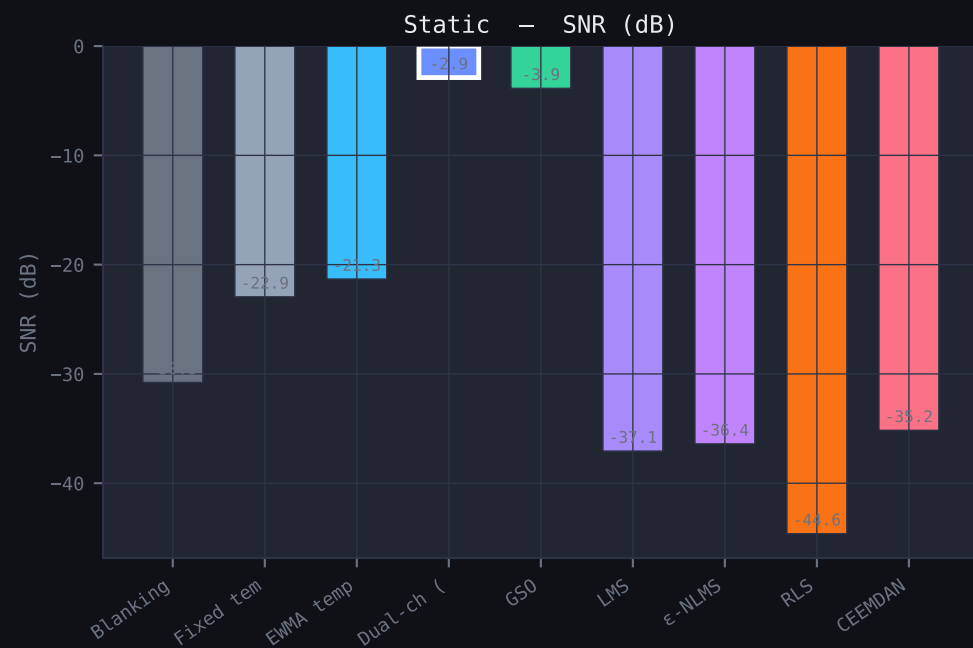
Algorithm colour key

 Blanking	 Fixed template	 EWMA template	 Dual-ch (DESTD)	 GS0
 LMS	 ϵ -NLMS	 RLS	 CEEMDAN	

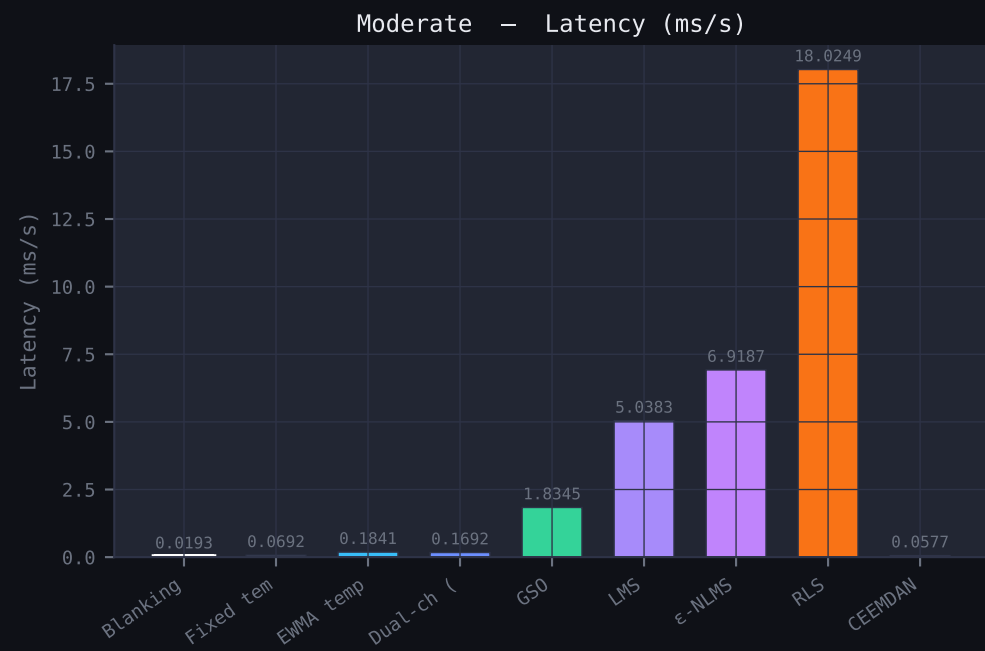
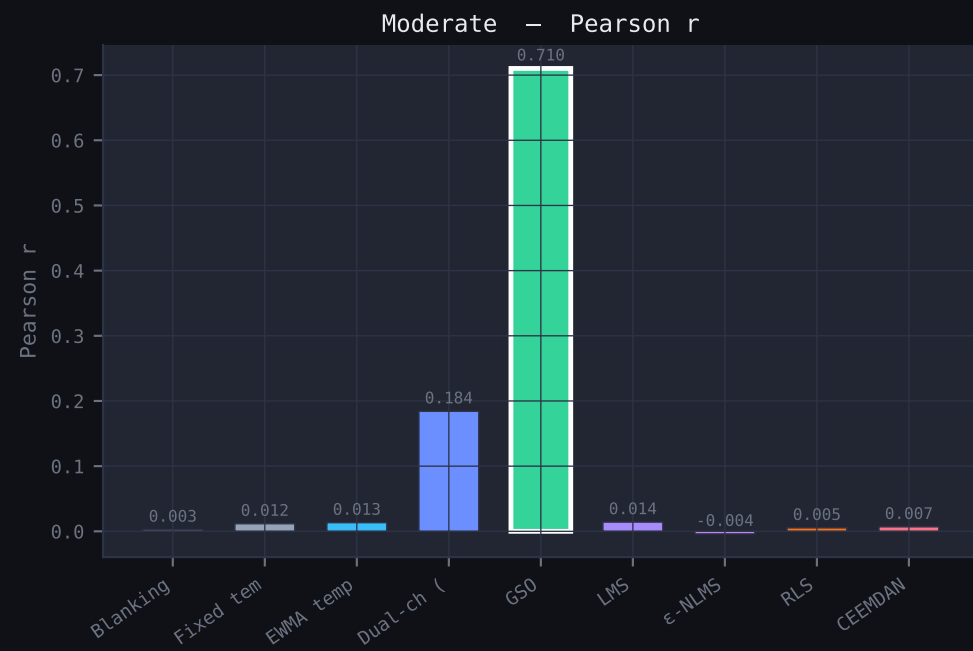
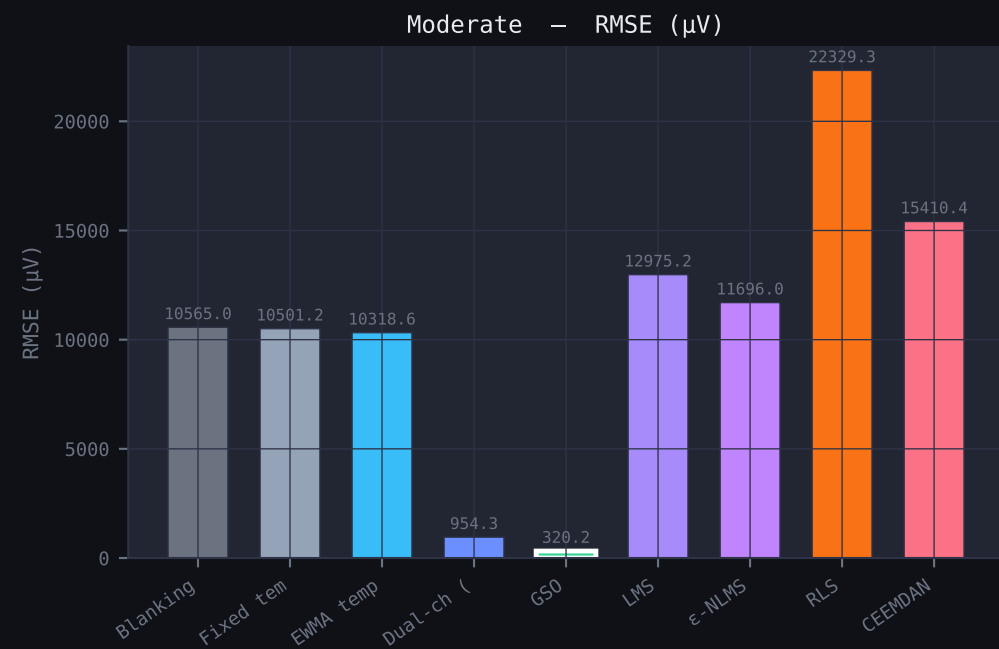
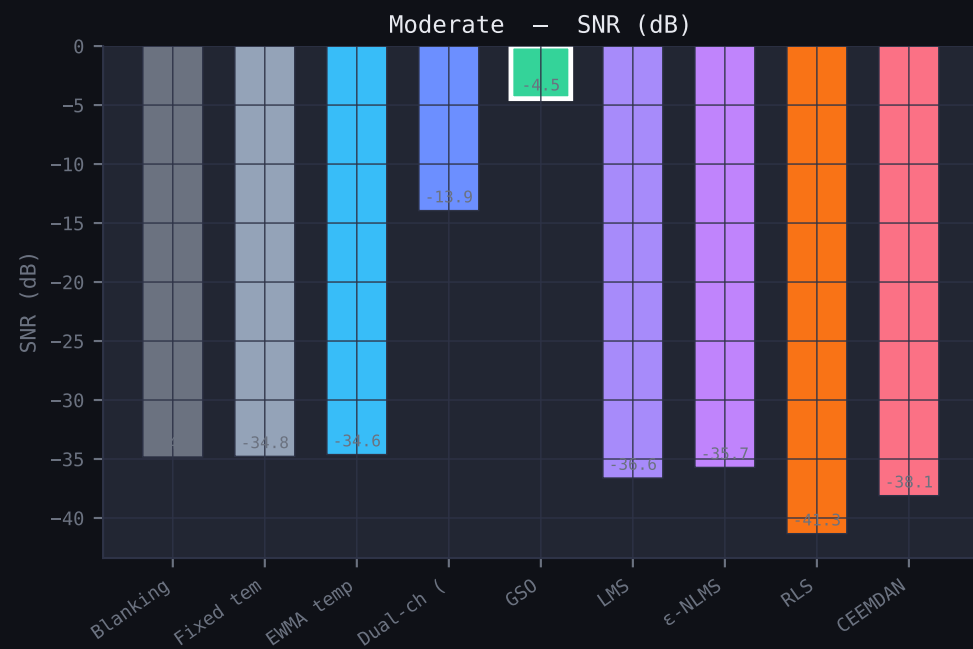
Contents

pp. 2–4	Bar charts (Static / Moderate / Stress)
p. 5	Static vs Stress grouped comparison
pp. 6–9	Degradation trajectories (SNR, RMSE, r , Latency)
p. 10	Quality trade-off scatter ($\text{RMSE} \times r$)
pp. 11–12	Latency ranking (Static & Stress)
p. 13	Delta heatmap: % change Static $\rightarrow$ Stress
pp. 14–15	Radar fingerprints (Static & Stress)

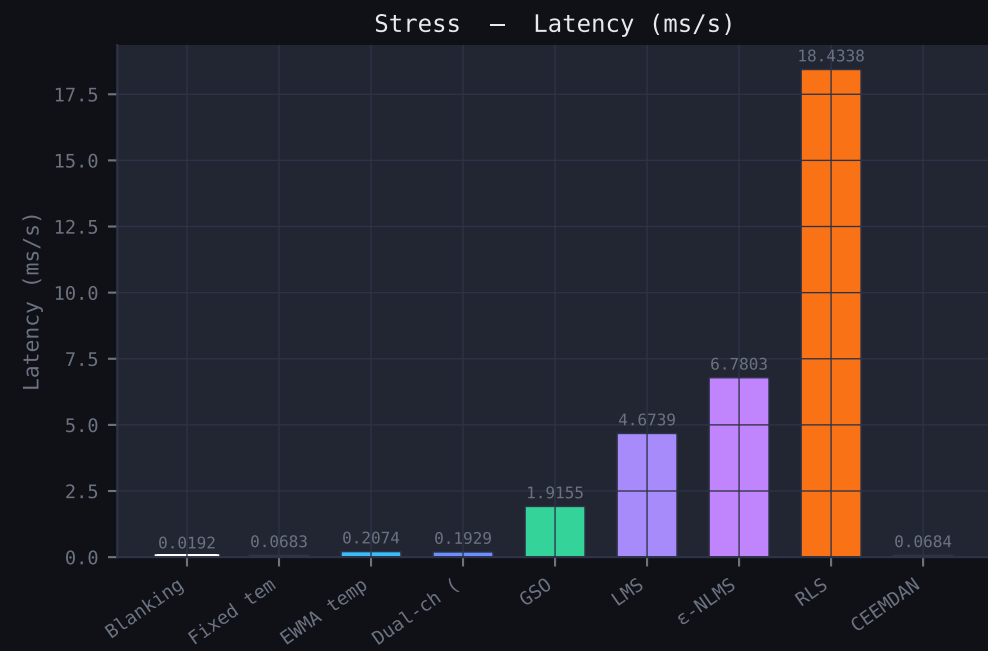
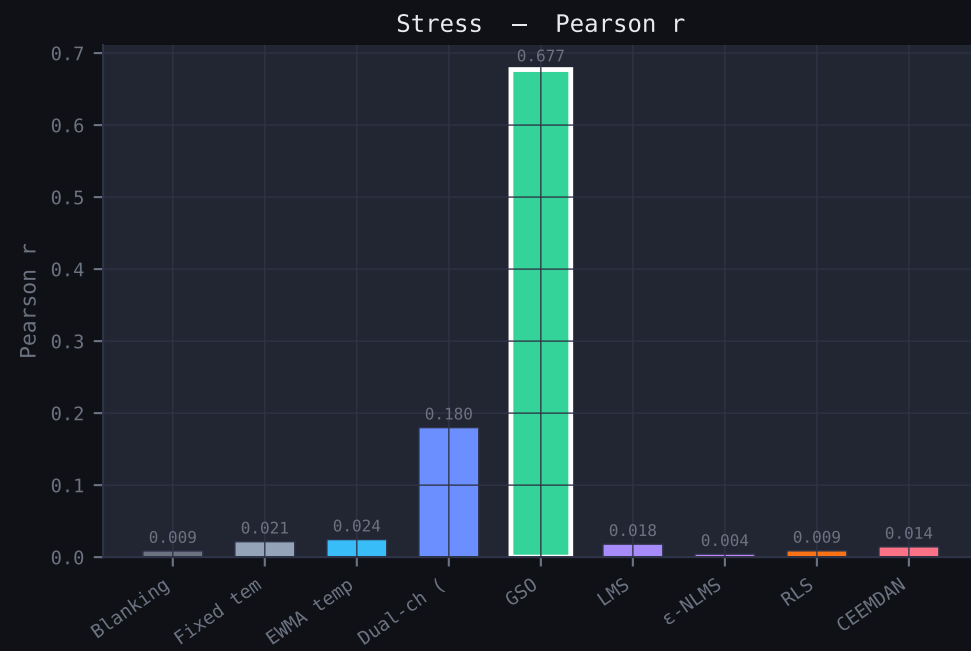
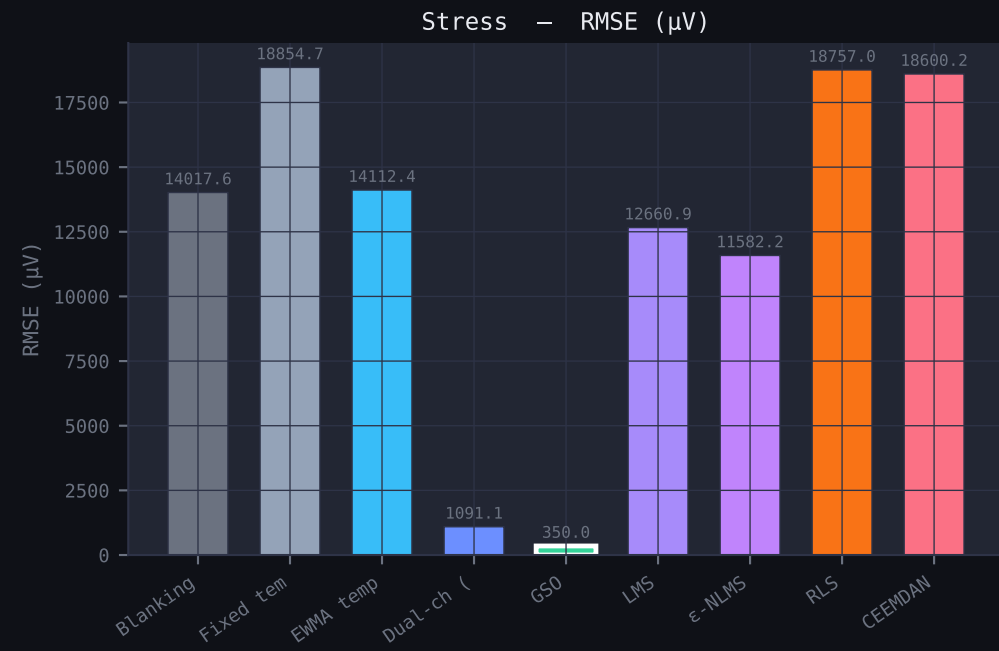
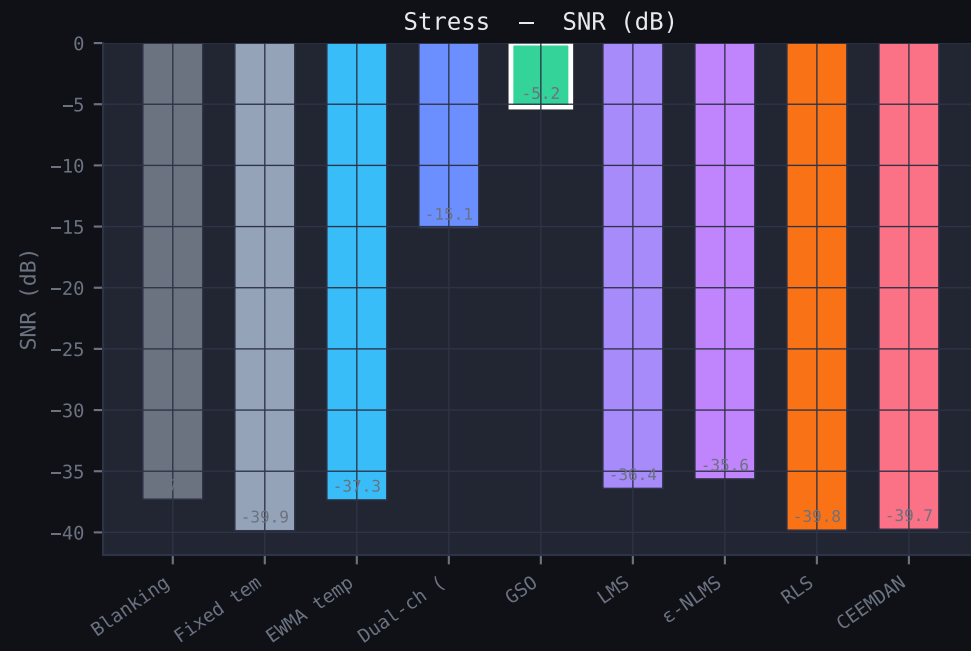
Algorithm performance – Static condition



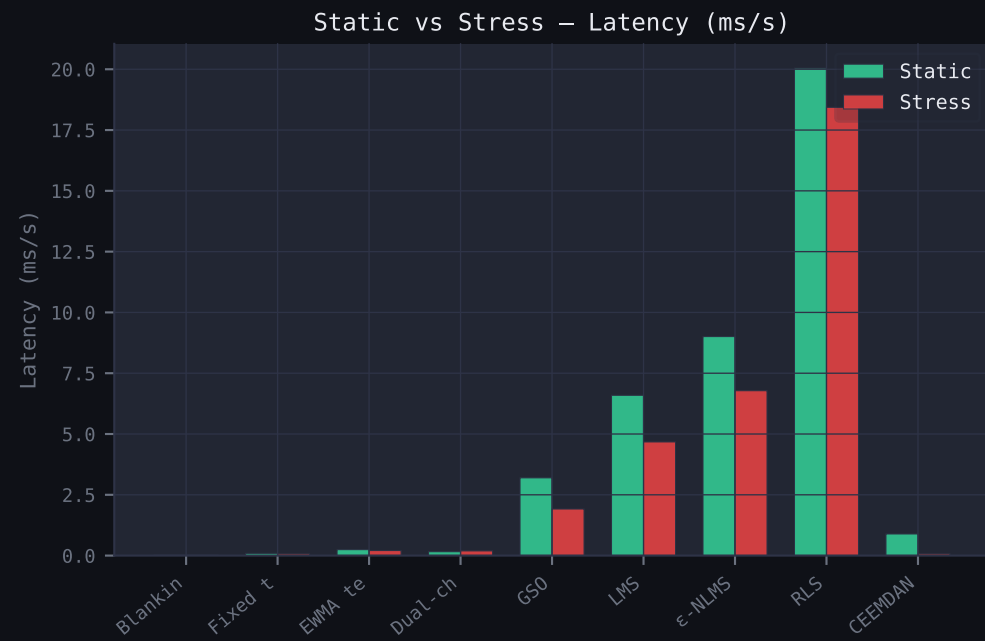
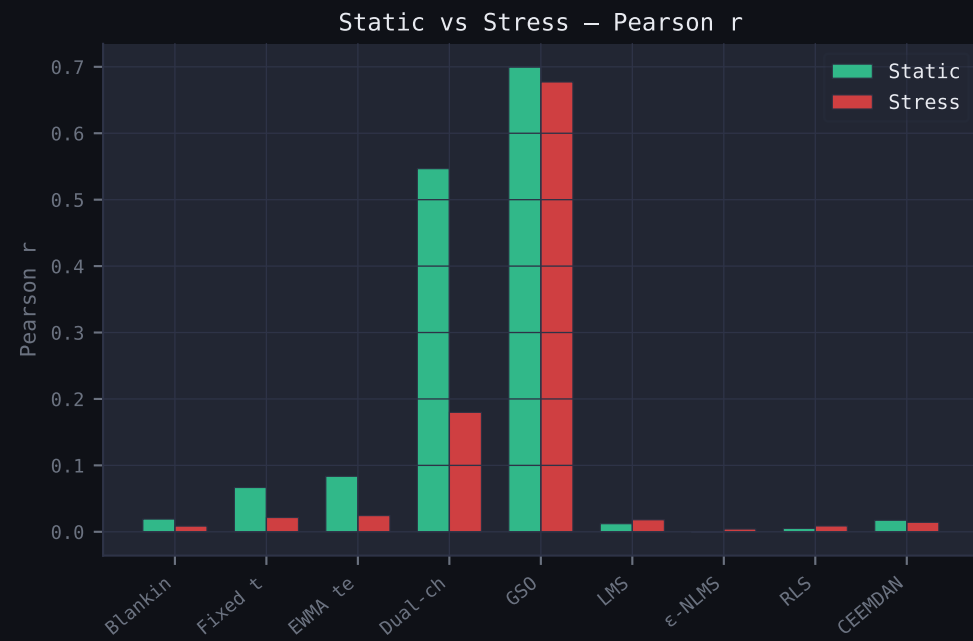
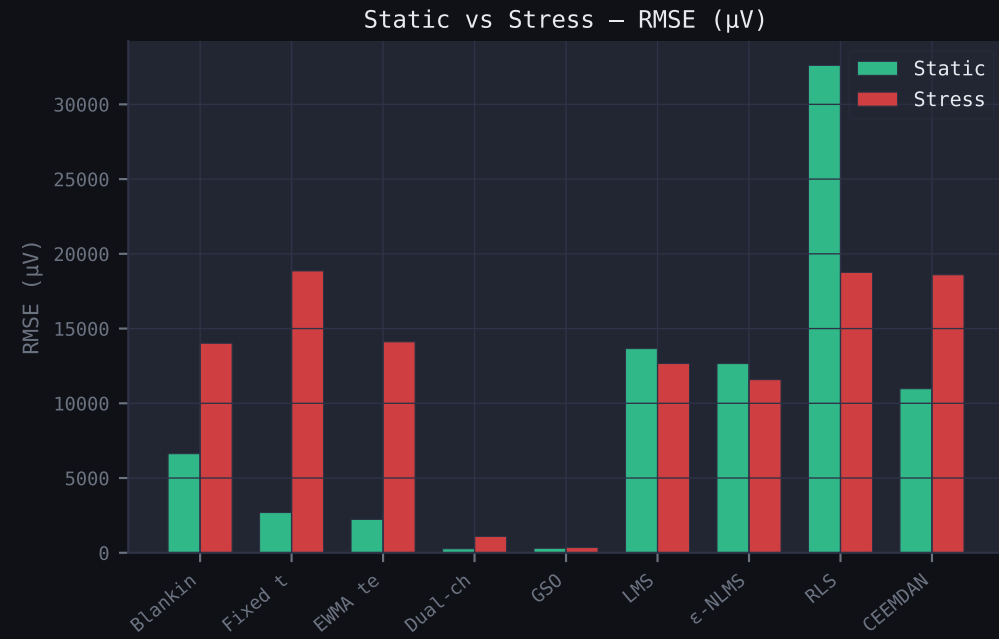
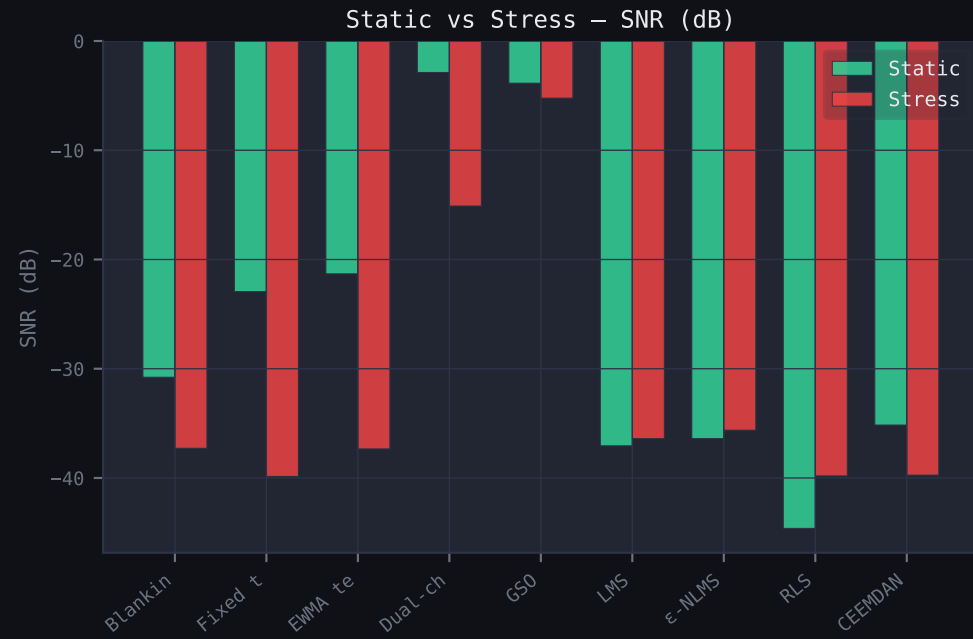
Algorithm performance – Moderate condition



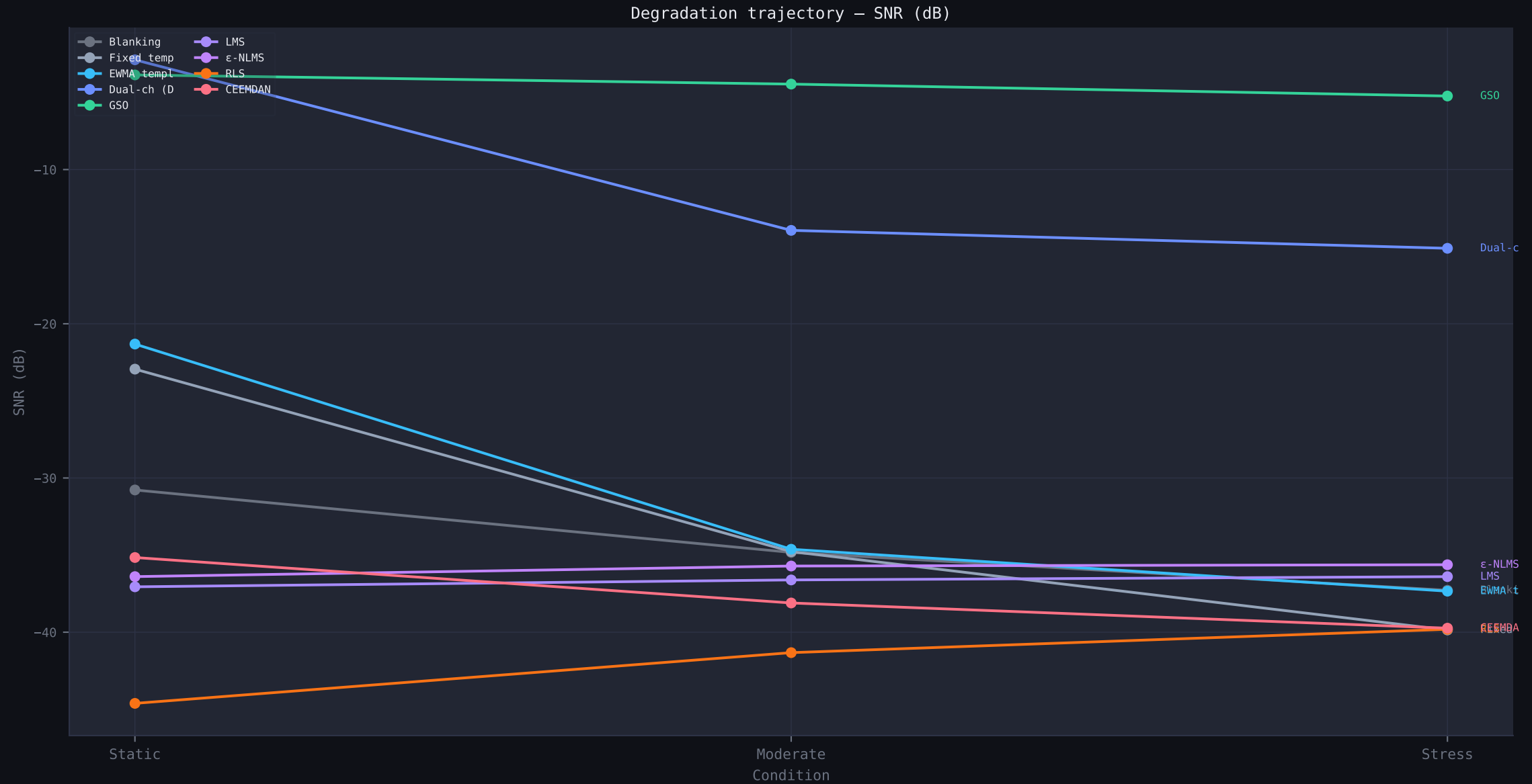
Algorithm performance – Stress condition



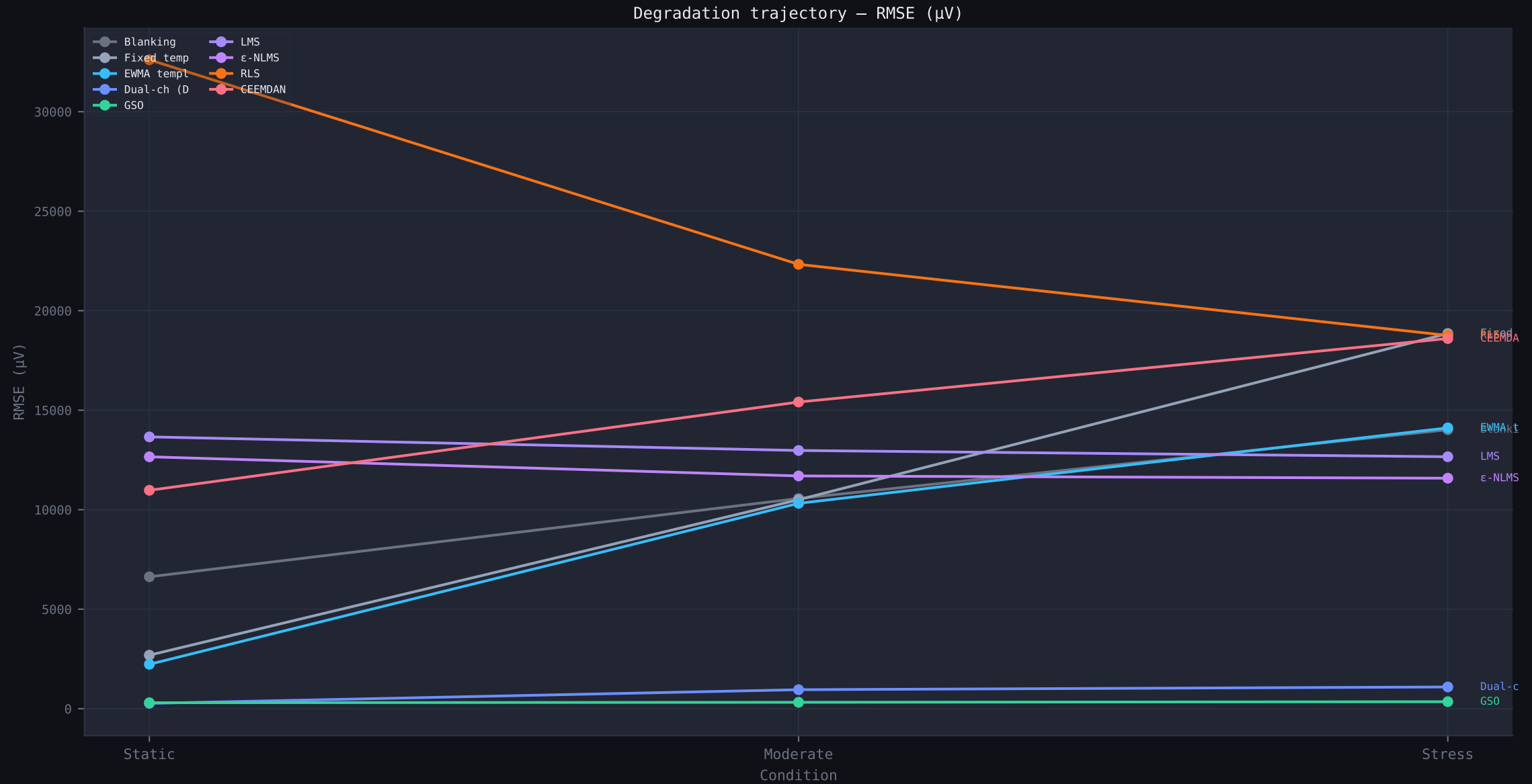
Static vs Stress – all metrics



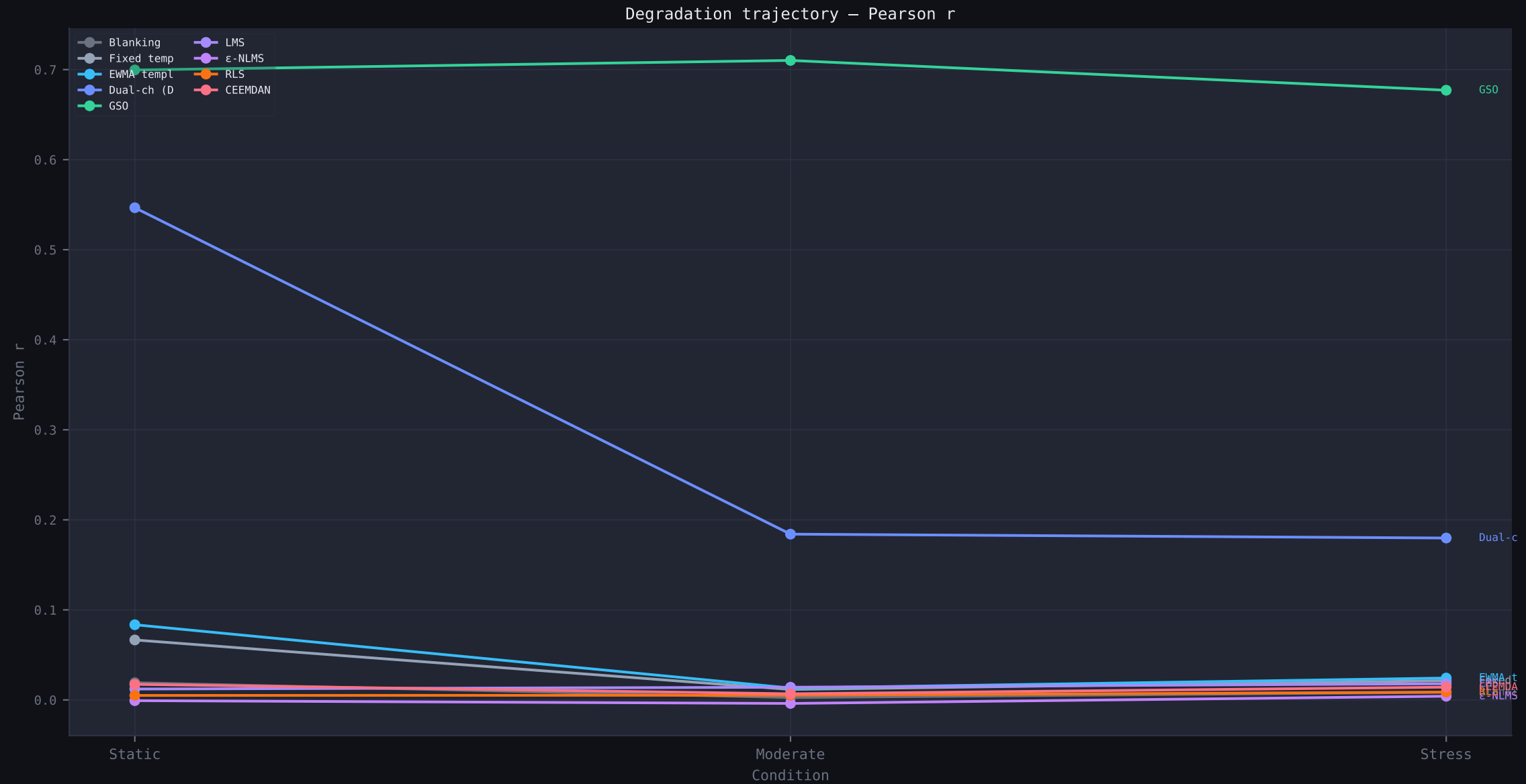
Degradation trajectory – SNR (dB)



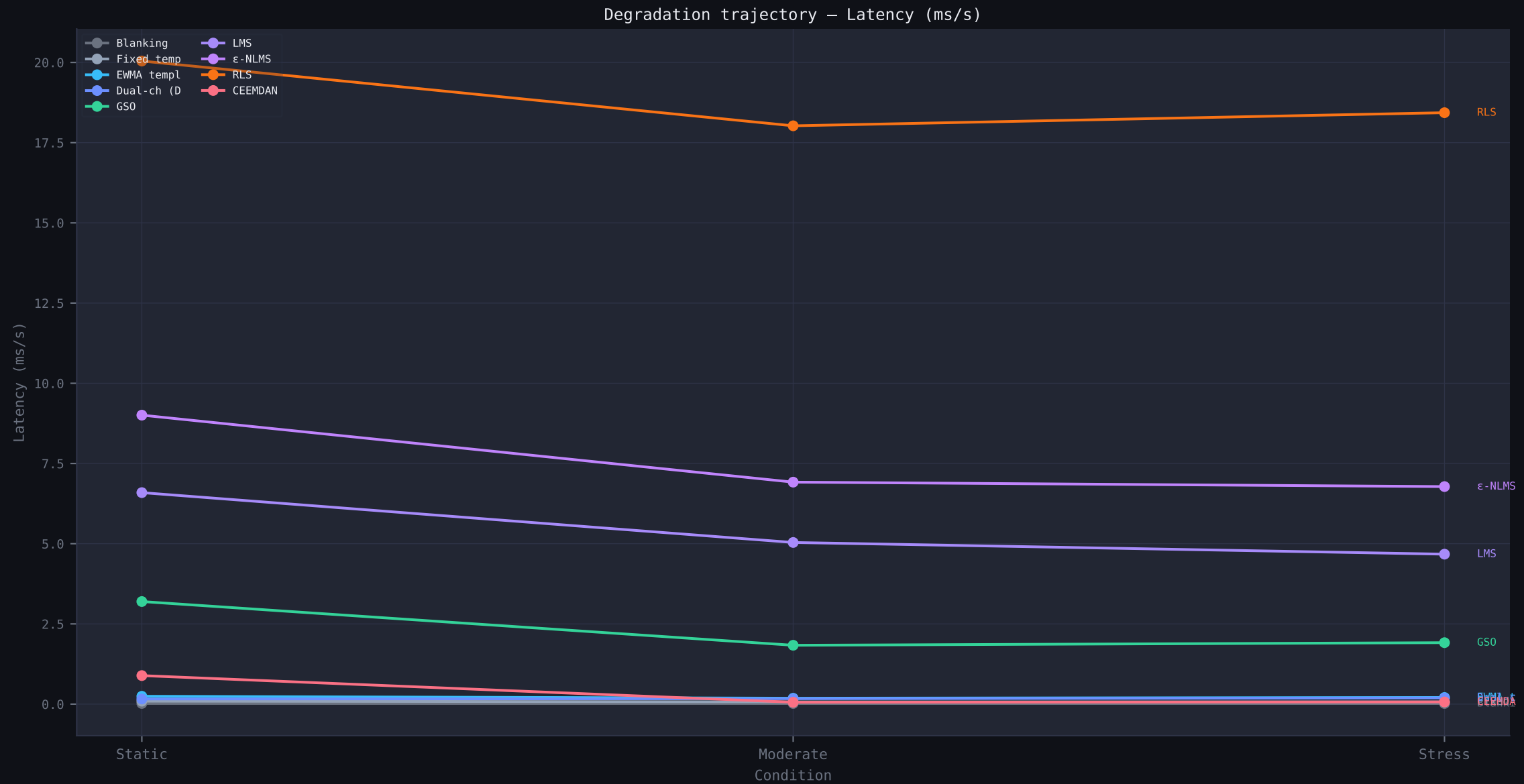
Degradation trajectory – RMSE (μV)



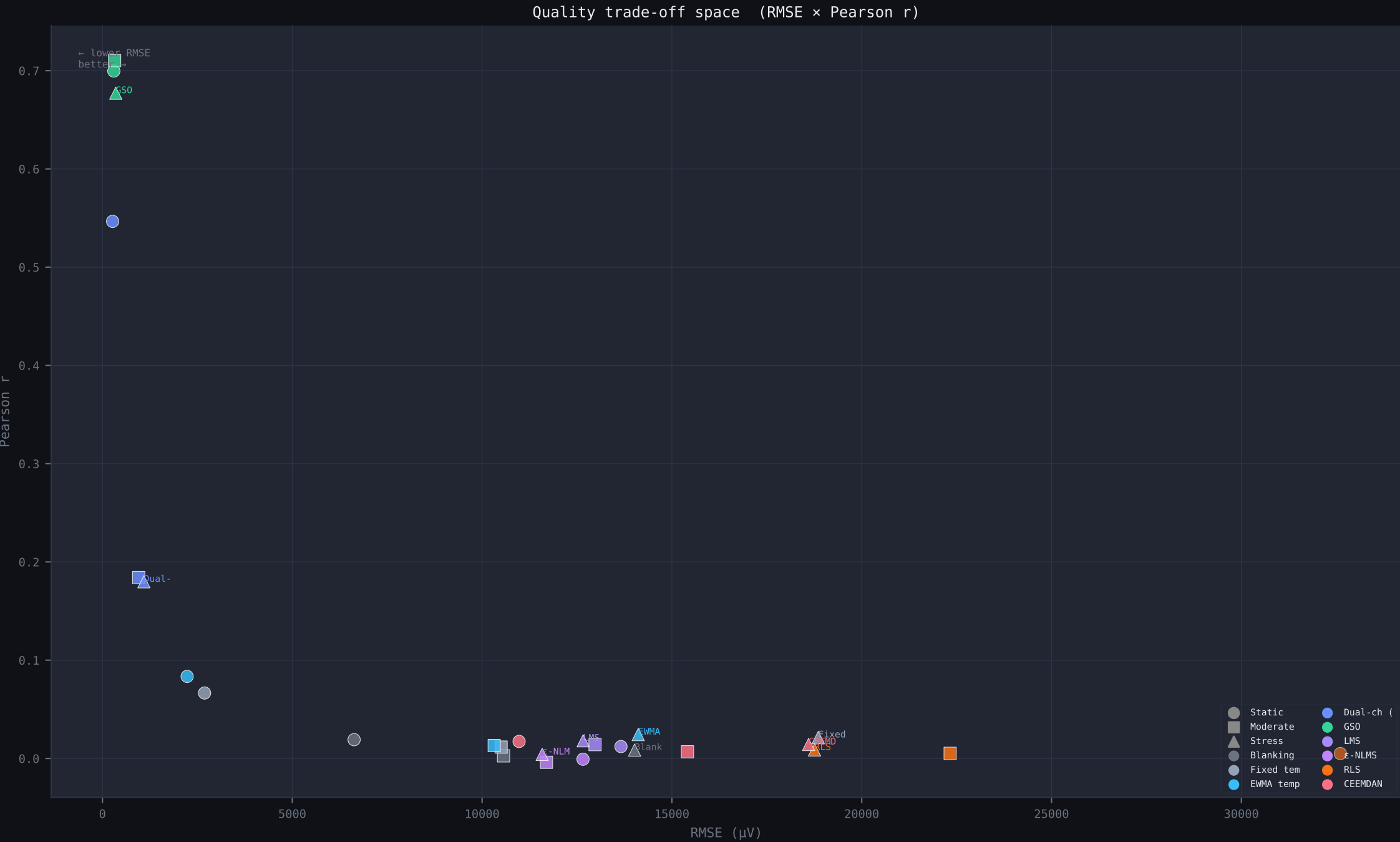
Degradation trajectory – Pearson r



Degradation trajectory – Latency (ms/s)

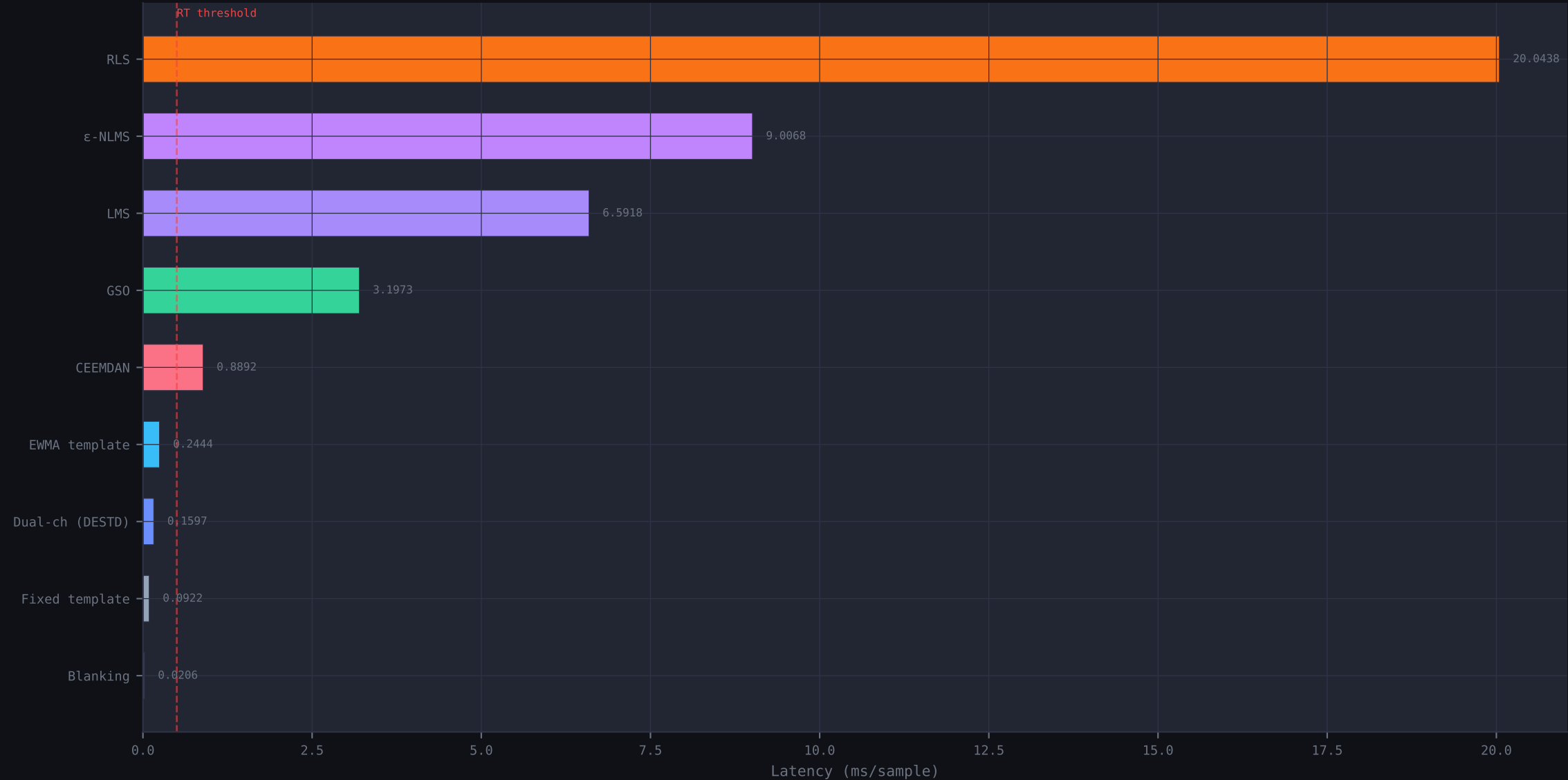


Quality trade-off space: RMSE × Pearson r



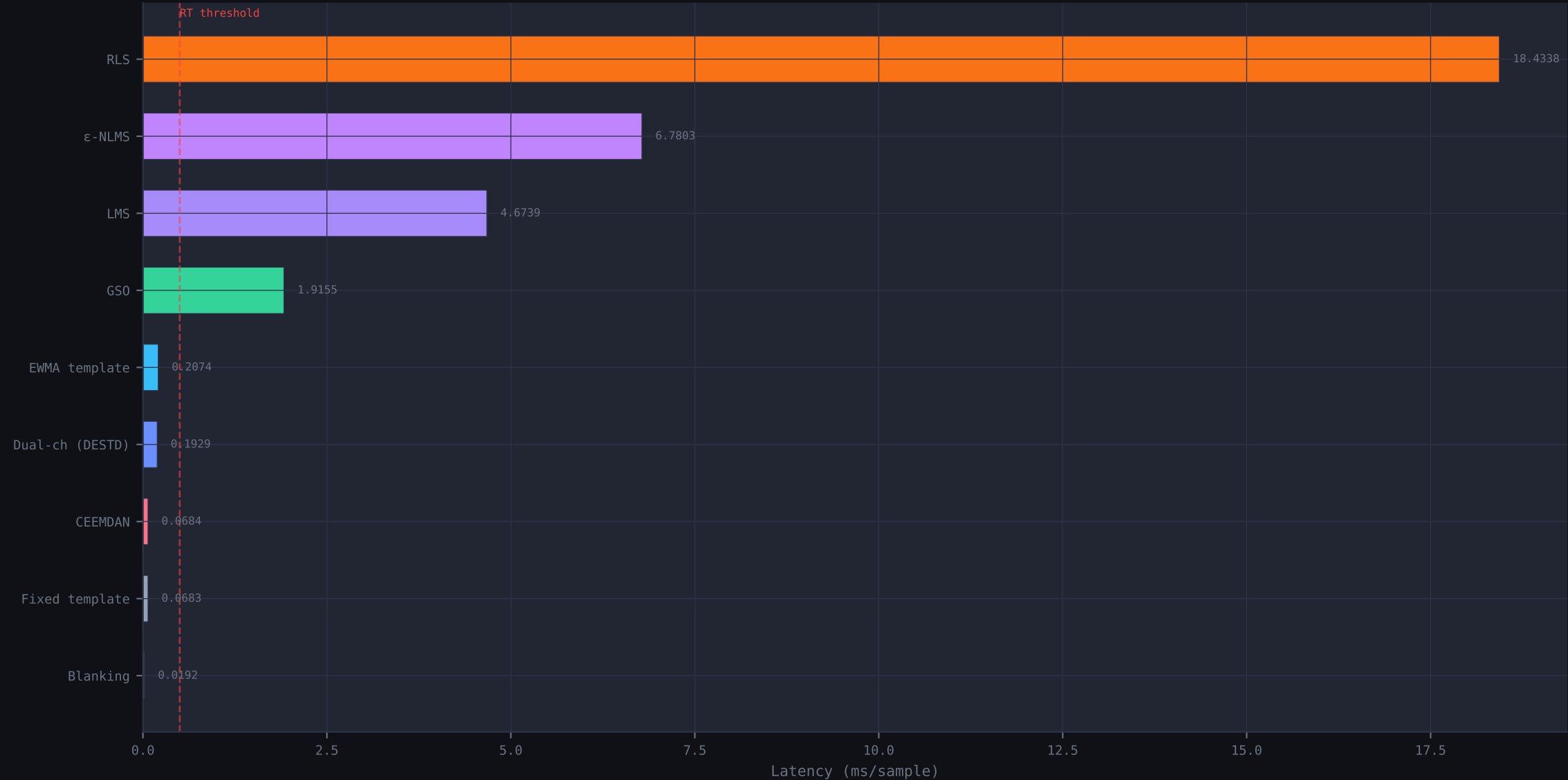
Latency ranking – Static

Latency – Static (ms per sample · lower = faster)



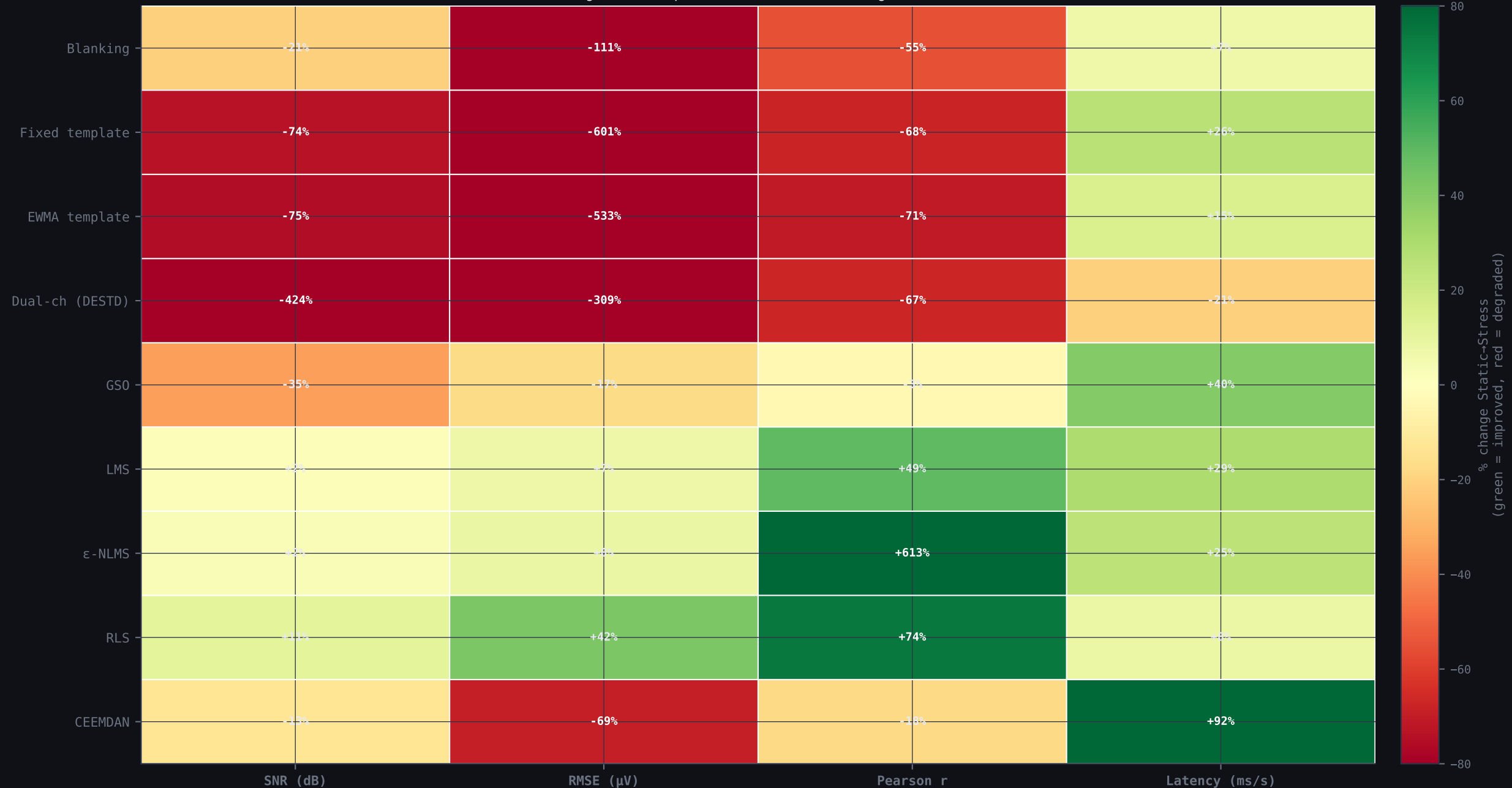
Latency ranking – Stress

Latency – Stress (ms per sample · lower = faster)



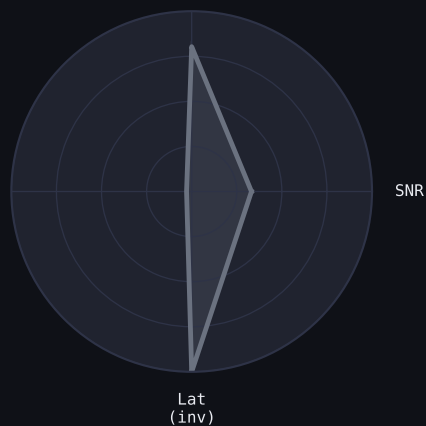
Robustness: % change Static → Stress

Robustness: % change from Static → Stress
(green = improved / held, red = degraded)



Multi-metric radar fingerprints – Static

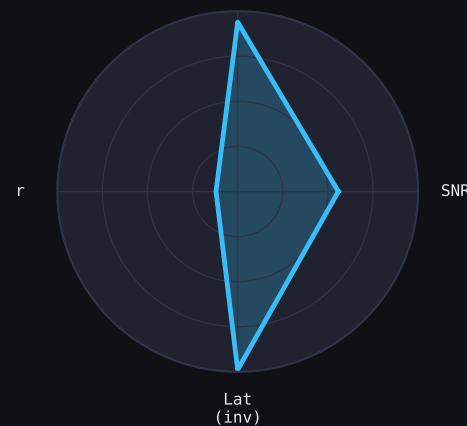
Blanking
RMSE
(inv)



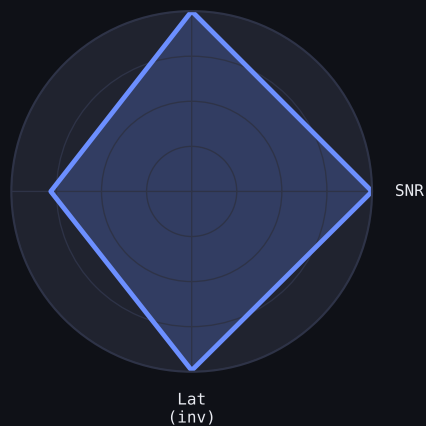
Fixed templ
RMSE
(inv)



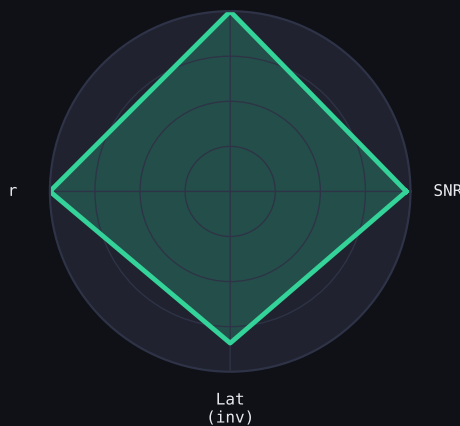
EWMA templa
RMSE
(inv)



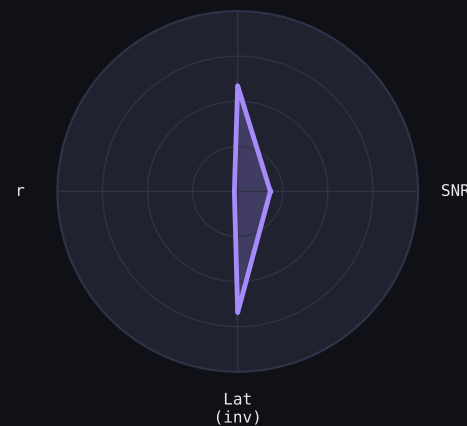
Dual-ch (DE
RMSE
(inv)



GS0
RMSE
(inv)



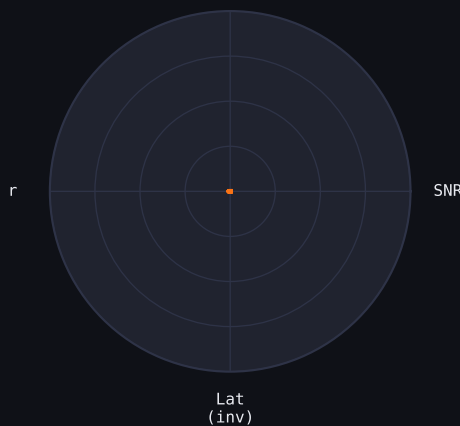
LMS
RMSE
(inv)



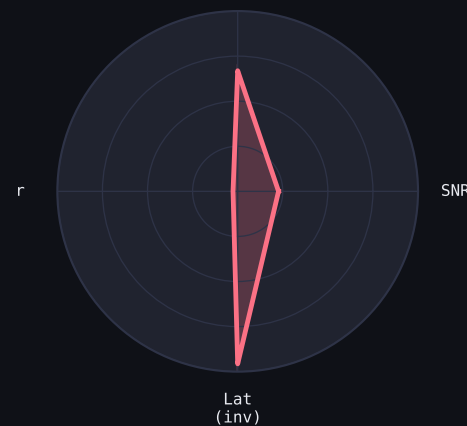
ϵ -NLMS
RMSE
(inv)



RLS
RMSE
(inv)

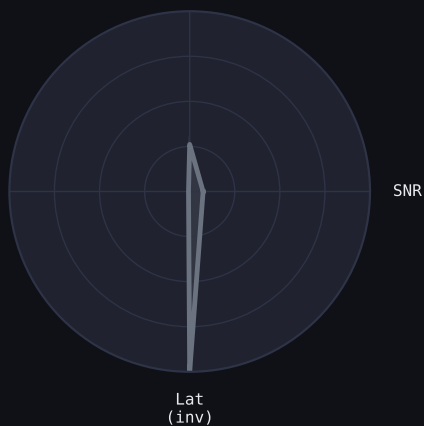


CEEMDAN
RMSE
(inv)

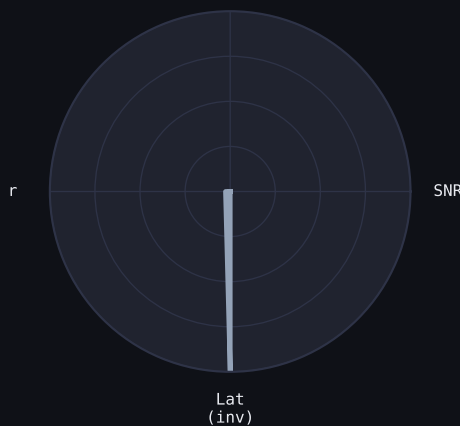


Multi-metric radar fingerprints – Stress

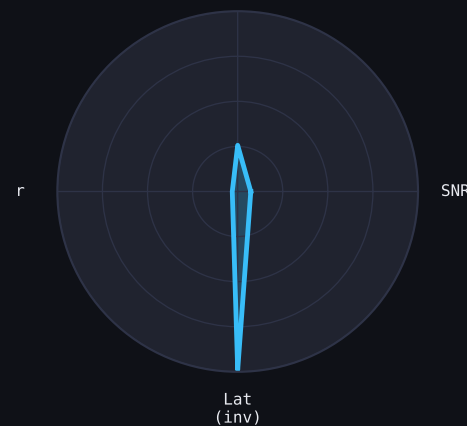
Blanking
RMSE
(inv)



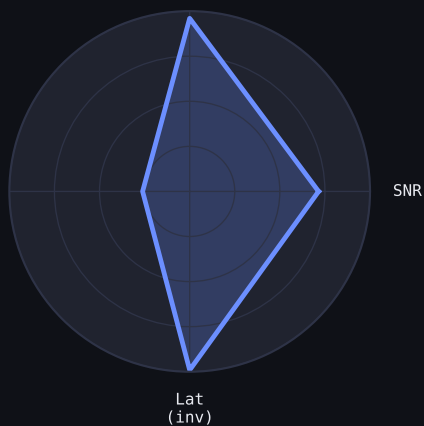
Fixed templ
RMSE
(inv)



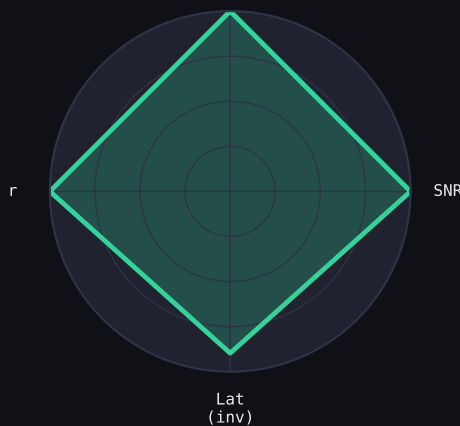
EWMA templa
RMSE
(inv)



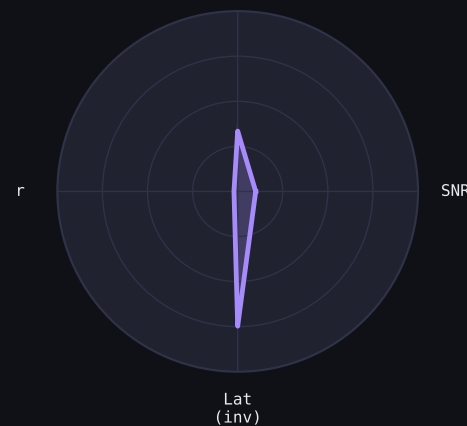
Dual-ch (DE
RMSE
(inv)



GS0
RMSE
(inv)



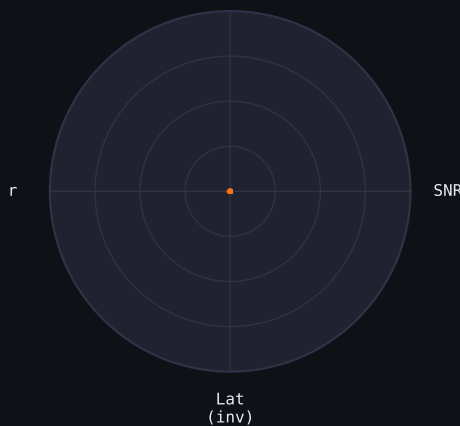
LMS
RMSE
(inv)



ϵ -NLMS
RMSE
(inv)



RLS
RMSE
(inv)



CEEMDAN
RMSE
(inv)

